

Question 1: Requirement and mini test plan

Part 1: Requirement Understanding

Task flow is a web based task management application. Users login with an account and do the task according to their role. For this application, there are two users one is a QA user and another one is a viewer. The QA user tests the application, which means they can login successfully and then add a task, and also can edit and delete the task created. Whereas, a viewer can only view, search and filter.

All testing is judged against the testing requirements.

Area	Req no	Key rule to test
Authentication	1-4	QA user credentials must allow login, invalid credentials must show a clear error and must not create a session, email login must ignore leading spaces and must be case-insensitive, logout must end the current session and return the user to the login screen
Roles	5-6	QA user can create, edit and delete tasks and the viewer can only view search and filter tasks but cannot create, edit or delete tasks
Tasks	7-14	Title is required and must be (3-50) characters, Description: must be (0-200) characters, Status: Todo/In Progress/Done, Priority: Low/Medium/High, Due date: Today or a future date, Editing and not duplicating, Confirmation for deletion of task, Showing the saved task in the list
Search and filter	15-17	Search must match with title and description being case-insensitive, filter must work, if not tasks match, user must see a clear empty-state message
Usability and accessibility	18-20	Must be responsive without any content becoming inaccessible, clear labels or names on the forms and keyboard focus should be visible on interactive controls

User journeys

ID	Journey	User flow	Req no
J1	Authentication	QA user sees the taskflow login page, user put the valid credentials, user logs in, user sees the lists of tasks, user logs out and return to the login page	1-4
J2	Create task	QA user clicks on the “Add Task” button, user make the title, description, choose status and priority and add due date and then save the task, user sees the task on the list	5,7-12
J3	Edit Task	QA user sees the existing task, click on “Edit”, make some changes in the details, user save the task, user sees the changes in the list	13
J4	Delete Task	QA user sees the existing task, click on “Delete” button, user sees a confirmation message, user says ok, task gets removed from the list	14
J5	Search and filter	User search the tasks according to the title or description, user sees the task searched, and empty if it does not exists	15,17
J6	Status filter	User chooses the options form dropdown and sees the tasks that are in those chosen status accordingly	16

Mini test plan

1. Objective

Verify the TaskFlow works/behaves according to the requirement or not.
Create test cases and then detect any bugs or issues in the application.
Record the bugs and give release recommendations. Also, automation flows in the playwright.

2. Scope

In scope

1. Authentication: User login with valid and invalid credentials, and empty fields
2. Roles: QA user and Viewer
3. Task Rules: QA can create, edit and delete the tasks but a viewer can only view, search and filter but not create, edit or delete the tasks
4. Search and filter: Search matching the title and description, status filter
5. Usability and accessibility: Main functions being usable at both the desktop and mobile widths, form fields and controls having clear labels or accessible names

Out of scope

1. Performance and load testing
2. Database and API testing
3. Security testing
4. Cross OS testing

3. Environment

Application URL	http://127.0.0.1:4173
Browser	Microsoft Edge
Operating system	Windows 11/Desktop
Tools	Google sheets, google docs, vs code, github

4. Approach

- a. Requirement based testing: Every expected result is taken from the written requirements and quotes the requirement number
- b. Smoke test first: A smoke checklist decides whether the build is stable enough for deeper testing
- c. Test types: Smoke, positive, negative, boundary, search-filter, responsiveness
- d. Execution order: Smoke check first, authentication, create, edit, delete, search, status-filter, role-permission, responsiveness and usability checks

- e. Defects: Logged in immediately with environment, expected and actual results, evidence, priority and severity
- f. Automation: locators, POM, assertions, test data

5. Entry and exit criteria

Entry Criteria

TaskFlow application is reachable
 QA user and viewer accounts are working
 Product requirements is available
 Playwright starter folder runs

Exit criteria

At least 15 test cases are written and and executed with results
 Smoke checks and responsive checks completed
 Bug report completed
 Automated tests run with HTML report

6. Deliverables

1. Mini test plan
2. Test scenarios and detailed test cases
3. Smoke check lists
4. Playwright project

7. Highest risk areas

Risk areas	What could go wrong	Why test it early
Authentication	Login fails and user can login successfully with invalid credentials, viewer may be able to create, edit or delete the task	Every other area depends on login. User will not be able to go through the application without successful login
Task creation	Title and description limits may not work, application may accept empty fields to create a task, application may accept yesterday;s date and not today or future dates.	Task creation is the core feature of this application. So , it is cheaper to catch early than affecting everything after it
Edit and delete task	Editing of a task may lead to create a duplicate data, no confirmation message on deletion of tasks	It can affect the user with duplication of data. These actions change or destroy the existing data. So, they have highest impact on QA user journey

